Tightness of the Longest Directed-Path Field in Two Peano Coordinates

Corrected proof with adversarial verification report

20 August 2026

Verification status. No genuinely external mathematical verifier was available through the connected tool environment when this document was produced. Consequently, this note does *not* claim external certification. It does contain (i) a source-by-source adversarial audit of the earlier argument, (ii) a corrected proof in which every probabilistic and topological passage is written out, and (iii) exhaustive finite-state checks of the KMSW interval identities and the zero-buffer BKR inequalities for all three-step words through length eight. The external-review limitation is recorded rather than concealed.

Abstract

Let M be the uniform infinite bipolar-oriented triangulation, equipped with its north-going and west-going Peano orders of the edges. These two orders give each edge a point of an infinite Baxter diagram. For two coordinatewise comparable edge-points, let D_n be $n^{-3/4}$ times the maximal number of edges in a directed path whose first and last edges are the specified edges. We prove tightness of the pairs consisting of the discrete comparable-pair domain and the graph of D_n in the local closed-set topology. More importantly, every subsequential limit is almost surely the graph of a locally bounded continuous function on its limiting domain.

The proof has a short mechanism. A consecutive packet of a longest directed path survives every resampling of KMSW letters outside the packet's exploration interval. Reimer's inequality therefore upgrades the known constant-order $m^{3/4}$ upper quantile for an m-step KMSW cell to an exponential upper tail. Applying this estimate in both Peano orders controls all narrow vertical and horizontal Baxter strips. A deterministic four-strip inequality then gives stochastic equicontinuity in all four endpoint coordinates. A variable-domain closed-graph compactness lemma completes the argument.

Contents

1 Statement, scope, and topology **2**

1.1 The two discrete Peano coordinates 2

1.2 Closed-set topology 3

2 Baxter rectangles and directed paths **4**

2.1 The BBMF Hasse diagram 4

2.2 Consecutive KMSW intervals 5

2.3 Chains confined to a strip 6

3 Exponential upper tails for KMSW blocks **6**
3.1 A zero-buffer BKR recursion 6
3.2 From one upper quantile to an exponential tail 7

4 Uniform estimates for all narrow strips **8**

5 The four-strip endpoint modulus **8**

6 Local height bounds **10**

7 A variable-domain closed-graph compactness theorem **10**

8 Proof of the main theorem **11**

9 Adversarial verification report **12**
9.1 Dependency audit 13

10 Exhaustive finite-state audit **14**

11 Conclusion **15**

1 Statement, scope, and topology

1.1 The two discrete Peano coordinates

Let M be the uniform infinite bipolar-oriented triangulation (UIBOT), with its edge orientations fixed. We use two bijective edge explorations

$$\lambda^{\mathrm{N}} : \mathbb{Z} \longrightarrow E(M), \qquad \lambda^{\mathrm{W}} : \mathbb{Z} \longrightarrow E(M),$$

called respectively the north-going and west-going Peano orders. The first is the usual KMSW order. We index the second so that increasing time is the exploration order of the dual down-right tree $T(m^*)$, i.e. the second coordinate in OP. The second is the primal west-going order of Gwynne–Holden–Sun (with the harmless deterministic time/sign convention needed for this indexing). Their Remark 1.8 states that its encoding walk is a bi-infinite i.i.d. three-step KMSW walk with the same law as the first encoding, after the deterministic sign convention used there; their Lemma 2.5 identifies this primal west-going order with the west-going order defined using the dual map [4]. Thus both coordinate directions admit the same KMSW block estimates. No independence between the two orders is asserted or needed.

For $e \in E(M)$, write

$$i(e) := (\lambda^{\mathrm{N}})^{-1}(e), \qquad j(e) := (\lambda^{\mathrm{W}})^{-1}(e), \qquad p(e) := (i(e), j(e)) \in \mathbb{Z}^2.$$

The finite-volume version of this pairing is exactly the Bonichon–Bousquet-Mélou–Fusy map OP: its first coordinate is the exploration order of the primal KMSW tree and its second coordinate is the exploration order of the corresponding dual tree [2, Definition 1.3 and Theorem 2.5]. The infinite version is obtained projectively from finite KMSW intervals; see [2, Sections 2.4 and 3.1].

For $n \in \mathbb{N}$, define the locally finite closed set

$$\mathcal{U}_n := \left\{ n^{-1} p(e) : e \in E(M) \right\} \subset \mathbb{R}^2$$

and the coordinatewise-comparable-pair domain

$$\mathcal{W}_n := \{(x, y) \in \mathcal{U}_n^2 : x_1 \le y_1,\ x_2 \le y_2\} \subset \mathbb{R}^4.$$

If $x = n^{-1}p(e)$ and $y = n^{-1}p(f)$ with $x \le y$, let

$$L_M(e, f)$$

be the maximal number of edges in a directed path of M whose first edge is e and whose last edge is f. The existence of such a path is proved in Lemma 2.1. Set

$$D_n(x, y) := n^{-3/4} L_M(e, f), \qquad (x, y) \in \mathcal{W}_n, \tag{1}$$

and

$$\Gamma_n := \mathrm{Graph}(\mathcal{W}_n, D_n) = \{(w, D_n(w)) : w \in \mathcal{W}_n\} \subset \mathbb{R}^5.$$

Since the first coordinates lie in $n^{-1}\mathbb{Z}$ with one edge per first coordinate, $\mathcal{U}_n$, $\mathcal{W}_n$, and Γ_n are locally finite and therefore closed. This is the edge-time formulation displayed on the blackboard. A vertex-corner formulation uses the neighboring white Hasse points and open rather than closed rectangles; the proof below is unchanged except for deterministic shifts of at most one lattice unit in each side coordinate.

1.2 Closed-set topology

For a locally compact metric space X, write $\mathsf{F}(X)$ for the set of closed subsets of X, including the empty set, equipped with the Fell topology. Since $X = \mathbb{R}^d$ is proper, this topology is metrizable and agrees with local Hausdorff convergence away from the boundaries of the truncating compact sets. Equivalently, one may identify $F \in \mathsf{F}(\mathbb{R}^d)$ with $F \cup \{\infty\}$ in the one-point compactification and use Hausdorff convergence there. In particular, $\mathsf{F}(\mathbb{R}^d)$ is compact.

We work in

$$\mathsf{F}(\mathbb{R}^4) \times \mathsf{F}(\mathbb{R}^5).$$

The mere closed-set tightness of Γ_n is therefore formal. The substantive assertion is that height does not escape over compact base sets and that no vertical segment survives in a limit.

Theorem 1.1 (Graph tightness and continuity)**.** *The laws of $(\mathcal{W}_n, \Gamma_n)$ are tight. Every subsequential weak limit $(\mathcal{W}, \Gamma)$ satisfies, almost surely,*

$$\Gamma = \{(w, D(w)) : w \in \mathcal{W}\}$$

for a unique locally bounded continuous function

$$D : \mathcal{W} \longrightarrow [0, \infty).$$

Corollary 1.2 (Identification after domain convergence)**.** *Suppose, along a subsequence, that the two-Peano meeting domains $\mathcal{W}_n$ converge in law to a specified random closed set $\mathcal{W}_\infty$. Then every joint subsequential limit of $(\mathcal{W}_n, \Gamma_n)$ is the graph of a continuous function on $\mathcal{W}_\infty$.*

Remark 1.3 (What is and is not proved here)**.** *The theorem proves the distance-field tightness and the continuous-graph property. It does not silently identify the limiting closed relation with*

$$\{(u, \widetilde{u}, v, \widetilde{v}) : \eta(u) = \widetilde{\eta}(\widetilde{u}),\ \eta(v) = \widetilde{\eta}(\widetilde{v}),\ u \le v,\ \widetilde{u} \le \widetilde{v}\}.$$

That identification is a separate convergence statement for the exact meeting-time relation of the two Peano curves. Uniform convergence of two curves alone does not imply Hausdorff convergence of their equality sets. Corollary 1.2 records precisely how any such domain theorem combines with the result proved here.

2 Baxter rectangles and directed paths

We first prove the exact deterministic correspondence needed later. It is convenient to state it for a finite plane bipolar orientation. The infinite statement follows by applying it to a finite KMSW interval and then using the projective consistency in [2, Proposition 2.4].

A finite KMSW interval is a marked bipolar orientation rather than necessarily a complete one. Complete its unexplored lower-left and upper-right boundary edges to obtain a finite bipolar orientation. The explored labels form one consecutive interval in the first OP coordinate, while every completion edge has first coordinate outside that interval. Hence a closed rectangle whose two black endpoints are explored contains no completion edge. The BBMF rectangle statement below therefore applies verbatim to explored endpoints and is independent of the chosen completion. Borga–Maazoun's projective construction of Θ and of the infinite permutation makes these finite patterns consistent.

2.1 The BBMF Hasse diagram

Let m be a finite plane bipolar orientation and let $\pi = \mathrm{OP}(m)$. In the Bonichon–Bousquet-Mélou–Fusy construction, the black points are

$$b_r = (r, \pi(r)), \qquad 1 \le r \le |E(m)|,$$

and the white points represent the vertices of m, together with the two pole sentinels. The strict product-order Hasse diagram on the black and white points is bipartite; every black point b has a unique lower white neighbor b^- and a unique upper white neighbor b^+; suppressing b turns the two Hasse edges

$$b^- \longrightarrow b \longrightarrow b^+$$

into the directed map edge represented by b [1, Proposition 1 and Theorem 2].

For black points $a \le b$, let

$$B[a,b] := \{c \text{ black} : a \le c \le b\}.$$

A chain always means a chain for the coordinatewise order.

Lemma 2.1 (Closed-rectangle path theorem)**.** *Let $a \le b$ be black points of the Baxter diagram of m, and let e_a, e_b be the corresponding map edges. Sending a directed path whose first edge is e_a and whose last edge is e_b to its sequence of black edge labels is a bijection onto the inclusion-maximal chains in $B[a,b]$ which contain a and b. Consequently,*

$$L_m(e_a, e_b) = \mathrm{LIS}(B[a,b]). \tag{2}$$

In particular, $a \le b$ if and only if there exists a directed path beginning with e_a and ending with e_b.

Proof. Let

$$e_{b_1}, e_{b_2}, \dots, e_{b_k}, \qquad b_1 = a,\ b_k = b,$$

be a directed path. If w_r is the common vertex between its rth and $(r+1)$st edges, then

$$b_r < w_r < b_{r+1}, \qquad w_r = b_r^+ = b_{r+1}^-.$$

Hence $b_1 < \cdots < b_k$ is a chain.

It is inclusion-maximal in $B[a,b]$. Indeed, suppose a black point c could be inserted strictly between b_r and b_{r+1}. Any saturated Hasse path from b_r to c starts with the unique upper-cover edge $b_r \to w_r$, so

$w_r \le c$. Any saturated Hasse path from c to b_{r+1} ends with the unique lower-cover edge $w_r \to b_{r+1}$, so $c \le w_r$. This is impossible because black and white points are distinct. There are no endpoint gaps because the chain already contains the minimum a and maximum b of the prescribed interval.

Conversely, let

$$a = b_1 < b_2 < \cdots < b_k = b$$

be an inclusion-maximal chain in $B[a, b]$. A saturated Hasse path from b_r to b_{r+1} has even length because the Hasse diagram is bipartite and both endpoints are black. If its length were at least four, it would contain an internal black point strictly between b_r and b_{r+1}, contradicting maximality. It therefore has length two, through one white point w_r. Uniqueness of the upper white neighbor of b_r and the lower white neighbor of b_{r+1} gives

$$w_r = b_r^+ = b_{r+1}^-.$$

Thus the map edges represented by $b_1, \ldots, b_k$ form a directed path, and the construction is inverse to the first map.

Every finite chain containing a, b extends to an inclusion-maximal such chain. Therefore the maximal path length equals the maximum cardinality of an arbitrary chain in $B[a, b]$, which is (2). □

Remark 2.2. *For comparable white points $u < v$, the analogous formula uses the black points in the open rectangle $(u_1, v_1) \times (u_2, v_2)$. The closed black-endpoint version is the one naturally matched to $\mathcal{U}_n$.*

2.2 Consecutive KMSW intervals

For the three-step triangulation encoding, write the KMSW letters as

$$X = (1, -1), \qquad L = (-1, 0), \qquad R = (0, 1).$$

A word of m letters produces active edges

$$\lambda_0, \lambda_1, \ldots, \lambda_m.$$

Every non-missing edge of the resulting marked triangulation appears exactly once in this list. The endpoint identities imply that exploration labels strictly increase along every directed path and that the entire incidence pattern among $\lambda_a, \ldots, \lambda_b$ depends only on the letters with indices $a+1, \ldots, b$. This is also the metric consequence of the restriction statement in [2, Proposition 2.4].

Lemma 2.3 (Interval isomorphism and zero-buffer persistence)**.** *Fix a KMSW word and integers $a \le b$. Run the KMSW procedure afresh on the subword with increment indices $a+1, \ldots, b$, beginning from one generic active edge. The correspondence*

$$\widehat{\lambda}_r \longmapsto \lambda_{a+r}, \qquad 0 \le r \le b - a,$$

extends to an isomorphism of edge-labelled directed incidence graphs.

Consequently, every directed path all of whose north-going labels lie in $[a, b]$ remains a directed path after arbitrary resampling of all KMSW letters outside $(a, b]$. The analogous statement holds for a west-going label interval.

Proof. For the north-going order, the endpoint equality tests are expressed solely through equalities and interval minima of the two KMSW walk coordinates and through the locations of the X-steps. Subtracting the walk value at time a leaves all of these tests unchanged. Hence every equality between

signed endpoints of the fresh edges $\widehat{\lambda}_r$ agrees with the corresponding equality between signed endpoints of λ_{a+r}. This proves the labelled incidence isomorphism. In particular, changing letters outside $(a, b]$ cannot alter any incidence between edges carrying labels in $[a, b]$.

For the west-going order, apply the identical argument to the west-going KMSW word. That word has i.i.d. increments with the same three-step law after the deterministic coordinate convention of [4, Remark 1.8]; its edge order agrees with the dual west-going order by [4, Lemma 2.5]. □

2.3 Chains confined to a strip

For a finite integer interval I, define

$$\Delta^{\mathrm{N}}(I) := \max\{|P| : P \text{ is a directed path and every edge of } P \text{ has north label in } I\},$$

and define $\Delta^{\mathrm{W}}(I)$ by replacing north with west. Here $|P|$ is the number of edges.

Lemma 2.4 (Strip completion)**.** *Let C be a chain of Baxter points. If every first coordinate of a point of C lies in an integer interval I, then*

$$|C| \le \Delta^{\mathrm{N}}(I).$$

If every second coordinate lies in I, then

$$|C| \le \Delta^{\mathrm{W}}(I).$$

Proof. Write $C = \{b_1 < \cdots < b_k\}$. Between each consecutive pair b_r, b_{r+1}, choose a saturated Hasse path. Every black point on such a path lies coordinatewise between b_r and b_{r+1}. Concatenating these saturated pieces and suppressing the black points produces a directed map path beginning with the edge b_1 and ending with the edge b_k. It contains all k prescribed edges. Every black label used by the path has first coordinate between the first coordinates of b_1 and b_k, hence in I. This proves the first assertion. The second is identical with the two coordinates interchanged. □

3 Exponential upper tails for KMSW blocks

Let Δ_m be the maximal length of a directed path in the marked triangulation produced by an i.i.d. three-step KMSW word of length m. There are $m + 1$ active edges, so $1 \le \Delta_m \le m + 1$.

3.1 A zero-buffer BKR recursion

For $r \in \mathbb{N}$, let

$$E_{m,r} := \{\Delta_m \ge r\}.$$

The probability space is the product space $\{X, L, R\}^m$ with the uniform product measure.

Lemma 3.1 (Zero-buffer disjoint occurrence)**.** *For every $m, k, r \in \mathbb{N}$,*

$$\mathbb{P}[\Delta_m \ge kr] \le \mathbb{P}[\Delta_m \ge r]^k. \tag{3}$$

Proof. On $\{\Delta_m \ge kr\}$, choose a directed path with at least kr edges and divide its first kr edges into k consecutive packets of r edges. For packet s, let a_s and b_s be the north-going labels of its first and last edges. Labels increase strictly along directed paths, so

$$a_1 \le b_1 < a_2 \le b_2 < \cdots < a_k \le b_k.$$

Set

$$K_s = (a_s, b_s] \cap \mathbb{Z} \subset \{1, \dots, m\}.$$

These coordinate sets are pairwise disjoint. By Lemma 2.3, fixing the KMSW letters in K_s preserves every incidence among the active edges with labels in $[a_s, b_s]$, no matter how all other letters are resampled. Hence the sth packet remains a directed path of length r. Thus K_s certifies $E_{m,r}$.

We have proved

$$E_{m,kr} \subset E_{m,r}^{\square k},$$

where the right side is k-fold disjoint occurrence. Reimer's inequality for arbitrary events in a finite product space gives

$$\mathbb{P}(E_{m,r}^{\square k}) \le \mathbb{P}(E_{m,r})^k,$$

which is (3). If $r = 1$ and a packet contains one edge, then K_s may be empty; this is harmless because $E_{m,1}$ is the sure event. □

Remark 3.2 (Why there is no two-edge loss)**.** *For shortest paths, resampling outside a block can create a shortcut, so one must discard endpoint buffers. A longest-path packet is an actual directed path and cannot be destroyed by adding or changing edges outside its labelled interval. This is the reason the present argument yields an exponential tail.*

3.2 From one upper quantile to an exponential tail

The only quantitative distance input is the upper half of Borga–Gwynne's KMSW-cell theorem: for every $\delta > 0$ there is $B(\delta) < \infty$ such that, uniformly in m,

$$\mathbb{P}[\Delta_m \le B(\delta)(m+1)^{3/4}] \ge 1 - \delta.$$

This is Theorem 1.12 of [3] (the replacement of m by $m+1$ only reflects the active-edge convention).

Proposition 3.3 (Uniform exponential block tail)**.** *There exist constants $c, C \in (0, \infty)$ such that, for every $m \ge 0$ and every $A \ge 0$,*

$$\mathbb{P}[\Delta_m > A(m+1)^{3/4}] \le Ce^{-cA}. \tag{4}$$

The same estimate holds for $\Delta^{\mathrm{N}}(I)$ and $\Delta^{\mathrm{W}}(I)$, with $m+1$ replaced by the cardinality of I.

Proof. Choose $B < \infty$ so that

$$\mathbb{P}[\Delta_m \le B(m+1)^{3/4}] \ge \frac{3}{4}$$

for every m. Let

$$r_m := \lfloor B(m+1)^{3/4} \rfloor + 1.$$

Then

$$\mathbb{P}[\Delta_m \ge r_m] \le \frac{1}{4}.$$

By Lemma 3.1,

$$\mathbb{P}[\Delta_m \ge k r_m] \le 4^{-k}, \qquad k \in \mathbb{N}.$$

Since $r_m \le (B+1)(m+1)^{3/4}$, for $A \ge 2(B+1)$ and

$$k = \left\lfloor \frac{A}{B+1} \right\rfloor$$

we have $k \geq A/[2(B+1)]$ and $kr_m \leq A(m+1)^{3/4}$. Therefore

$$\mathbb{P}[\Delta_m > A(m+1)^{3/4}] \leq 4^{-k} \leq \exp\left(-\frac{\log 4}{2(B+1)}A\right).$$

Increasing the multiplicative constant handles bounded A.

For a north label interval I, Lemma 2.3 identifies the induced directed incidence graph with a fresh KMSW block containing $|I|$ active labels. The west-going statement follows from the west-going KMSW encoding discussed in Section 1.1. Hence the same constants apply. □

4 Uniform estimates for all narrow strips

For $R > 0$, $\delta > 0$, and $\star \in \{\mathrm{N}, \mathrm{W}\}$, define

$$\Xi^{\star}_{n,R}(\delta) := n^{-3/4} \sup\{\Delta^{\star}(I) : I \subset [-Rn, Rn] \cap \mathbb{Z} \text{ an interval, } |I| \leq \delta n + 3\}. \tag{5}$$

Proposition 4.1 (Uniform narrow-strip estimate)**.** *For every $R, \varepsilon > 0$ and $\star \in \{\mathrm{N}, \mathrm{W}\}$,*

$$\lim_{\delta \downarrow 0} \limsup_{n \to \infty} \mathbb{P}[\Xi^{\star}_{n,R}(\delta) > \varepsilon] = 0. \tag{6}$$

More precisely, for all sufficiently large n depending on δ,

$$\mathbb{P}[\Xi^{\star}_{n,R}(\delta) > \varepsilon] \leq C_R(1+\delta^{-1}) \exp(-c\varepsilon\delta^{-3/4}). \tag{7}$$

Proof. Set $h = \lceil \delta n + 3 \rceil$ and consider the deterministic intervals

$$J_q = [qh, (q+2)h] \cap \mathbb{Z}, \qquad q \in \mathbb{Z}.$$

Every integer interval of cardinality at most h is contained in one of the J_q. At most $C_R(1+\delta^{-1})$ of these grid intervals meet $[-Rn, Rn]$. Each relevant J_q has cardinality at most $2h + 1 \leq C\delta n$ when n is sufficiently large in terms of δ.

Proposition 3.3 gives

$$\mathbb{P}[\Delta^{\star}(J_q) > \varepsilon n^{3/4}] \leq C \exp\left[-c\frac{\varepsilon n^{3/4}}{|J_q|^{3/4}}\right] \leq C \exp(-c'\varepsilon\delta^{-3/4}).$$

A union bound proves (7). The right side tends to zero as $\delta \downarrow 0$, proving (6). □

5 The four-strip endpoint modulus

For raw lattice points $x = (a, c) \leq y = (b, d)$, put

$$R[x, y] := [a, b] \times [c, d].$$

By Lemma 2.1, the unscaled distance represented by (x, y) is the longest-chain cardinality in the set of Baxter points lying in $R[x, y]$.

Lemma 5.1 (Four-strip inequality)**.** *Let*

$$w = (x, y), \qquad w' = (x', y')$$

be two comparable edge pairs, where

$$x = (a, c), \quad y = (b, d), \quad x' = (a', c'), \quad y' = (b', d').$$

Define the side intervals

$$I_a = [a \wedge a', a \vee a'], \quad I_b = [b \wedge b', b \vee b'],$$
$$J_c = [c \wedge c', c \vee c'], \quad J_d = [d \wedge d', d \vee d'].$$

Then

$$\begin{aligned} |L(w) - L(w')| \le \Delta^{\mathrm{N}}(I_a) + \Delta^{\mathrm{N}}(I_b) \\ + \Delta^{\mathrm{W}}(J_c) + \Delta^{\mathrm{W}}(J_d), \end{aligned} \tag{8}$$

where L denotes the unscaled longest directed-path value.

Proof. Let C be a longest chain in $R[x, y]$. The set $C \cap R[x', y']$ is a chain in the second rectangle, so

$$|C \cap R[x', y']| \le L(w').$$

Every point $z = (z_1, z_2)$ in $R[x, y] \setminus R[x', y']$ belongs to at least one of the four strips

$$z_1 \in I_a, \qquad z_1 \in I_b, \qquad z_2 \in J_c, \qquad z_2 \in J_d.$$

Indeed, if $z_1 \notin [a', b']$, then, because $z_1 \in [a, b]$, it lies between the two left sides or between the two right sides. If $z_1 \in [a', b']$, then $z \notin R[x', y']$ forces the analogous alternative for z_2.

The points of C in each strip form a chain. Lemma 2.4 bounds the cardinality of each such subchain by the corresponding Δ term. Hence

$$L(w) \le L(w') + \Delta^{\mathrm{N}}(I_a) + \Delta^{\mathrm{N}}(I_b) + \Delta^{\mathrm{W}}(J_c) + \Delta^{\mathrm{W}}(J_d).$$

Interchanging w and w' proves (8). □

For $R, \delta > 0$, let $K_R = [-R, R]^4$ and define

$$\omega_{n,R}(\delta) := \sup \left\{ |D_n(w) - D_n(w')| : w, w' \in \mathcal{W}_n \cap K_R, \ \|w - w'\|_\infty \le \delta \right\}. \tag{9}$$

Proposition 5.2 (Stochastic equicontinuity)**.** *For every $R, \varepsilon > 0$,*

$$\lim_{\delta \downarrow 0} \limsup_{n \to \infty} \mathbb{P}[\omega_{n,R}(\delta) > \varepsilon] = 0. \tag{10}$$

Proof. If $\|w - w'\|_\infty \le \delta$, each of the four raw lattice side intervals in Lemma 5.1 has cardinality at most $\delta n + 2$. All of them lie in $[-(R+1)n, (R+1)n]$ once n is large. Therefore

$$\omega_{n,R}(\delta) \le 2\Xi^{\mathrm{N}}_{n,R+1}(\delta) + 2\Xi^{\mathrm{W}}_{n,R+1}(\delta),$$

up to replacing δ by $\delta + 3/n$, which does not affect the iterated limit. Proposition 4.1 proves (10). □

6 Local height bounds

Define

$$H_{n,R} := \sup\{D_n(w) : w \in \mathcal{W}_n \cap K_R\},$$

with the supremum of the empty set equal to zero.

Proposition 6.1 (Compact-base height tail)**.** *For each $R > 0$, there are $c_R, C_R > 0$ such that*

$$\sup_{n\geq 1} \mathbb{P}[H_{n,R} > A] \leq C_R e^{-c_R A}, \qquad A \geq 0. \tag{11}$$

In particular, $\{H_{n,R} : n \geq 1\}$ is tight.

Proof. If $w = (x, y) \in \mathcal{W}_n \cap K_R$, every Baxter point in the rectangle $R[nx, ny]$ has north label in

$$I_{n,R} := [-Rn - 1, Rn + 1] \cap \mathbb{Z}.$$

Lemma 2.4 therefore gives

$$D_n(w) \leq n^{-3/4} \Delta^{\mathrm{N}}(I_{n,R}).$$

The interval has cardinality at most $C_R n$. Proposition 3.3 yields (11), after adjusting constants for finitely many small n. □

7 A variable-domain closed-graph compactness theorem

The topological step is deterministic. Writing it separately prevents the two errors in the earlier draft: tightness of local suprema does not by itself give almost-sure boundedness on a Skorokhod coupling, and an informal claim that an oscillation event is open does not replace a compactness argument.

Theorem 7.1 (Variable-domain Arzelà–Ascoli)**.** *For each $r, k \in \mathbb{N}$, let $A_r < \infty$ and $\delta_{r,k} > 0$, with $\delta_{r,k} \downarrow 0$ as $k \to \infty$. Consider pairs (W, G) for which*

$$W \in \mathsf{F}(\mathbb{R}^4), \qquad G = \mathrm{Graph}(W, f) \in \mathsf{F}(\mathbb{R}^5)$$

for some $f : W \to [0, \infty)$, and which satisfy

$$\sup_{x \in W \cap K_r} f(x) \leq A_r, \tag{12}$$

$$|f(x) - f(y)| \leq k^{-1} \quad \text{whenever } x, y \in W \cap K_r,\ \|x - y\|_\infty \leq 2\delta_{r,k}. \tag{13}$$

The closure of this family in

$$\mathsf{F}(\mathbb{R}^4) \times \mathsf{F}(\mathbb{R}^5)$$

is compact and every element of the closure is the graph of a locally bounded continuous function on its first component.

Proof. Compactness of the ambient product Fell hyperspace gives a convergent subsequence from every sequence in the family. Let

$$(W_j, G_j) \longrightarrow (W, G), \qquad G_j = \mathrm{Graph}(W_j, f_j).$$

We prove that G is a continuous graph over W.

Let $\pi : \mathbb{R}^5 \to \mathbb{R}^4$ be the base projection. If $(x,t) \in G$, the defining lower-limit property of Fell convergence provides $(x_j, f_j(x_j)) \in G_j$ converging to (x,t). Since $x_j \in W_j$ and $W_j \to W$, we have $x \in W$. Thus

$$\pi(G) \subset W.$$

Conversely, fix $x \in W$ and choose r so that x lies in the interior of K_r. There are $x_j \in W_j$ with $x_j \to x$. By (12), $f_j(x_j) \le A_r$ for all large j. Passing to a subsequence gives $f_j(x_j) \to t$, and then $(x,t) \in G$. Hence

$$\pi(G) = W.$$

Suppose $(x,s),(x,t) \in G$. Choose approximating graph points

$$(x_j, f_j(x_j)) \to (x,s), \qquad (y_j, f_j(y_j)) \to (x,t).$$

Fix r with x in the interior of K_r and fix k. For all sufficiently large j, both base points belong to K_r and

$$\|x_j - y_j\|_\infty \le 2\delta_{r,k}.$$

By (13),

$$|f_j(x_j) - f_j(y_j)| \le k^{-1}.$$

Letting $j \to \infty$, then $k \to \infty$, gives $s = t$. Thus every vertical fiber is a singleton, and

$$G = \mathrm{Graph}(W, f)$$

for a unique function f.

The limit inherits a modulus. If $x, y \in W$ lie in the interior of K_r and

$$\|x - y\|_\infty < \delta_{r,k},$$

choose $x_j, y_j \in W_j$ converging respectively to x, y. For large j their distance is at most $2\delta_{r,k}$, so (13) and the singleton-fiber conclusion imply

$$|f(x) - f(y)| \le k^{-1}.$$

Given $x_m \to x$ in W, choose r containing the sequence locally. For any $\varepsilon > 0$, take k with $k^{-1} < \varepsilon$; then the last estimate applies for all sufficiently large m. Hence $f(x_m) \to f(x)$. Local boundedness follows from (12). This completes the proof. □

8 Proof of the main theorem

Proof of Theorem 1.1. Fix $\eta > 0$. By Proposition 6.1, choose deterministic $A_r < \infty$ such that

$$\sup_n \mathbb{P}[H_{n,r} > A_r] \le \eta 2^{-r-2}, \qquad r \in \mathbb{N}. \tag{14}$$

Next fix $r, k \in \mathbb{N}$. Proposition 5.2 lets us choose $\delta^0_{r,k} > 0$ such that

$$\limsup_{n\to\infty} \mathbb{P}[\omega_{n,r}(2\delta^0_{r,k}) > k^{-1}] \le \eta 2^{-r-k-4}.$$

Hence the same probability is at most $\eta 2^{-r-k-3}$ for all $n \ge N_{r,k}$, for some deterministic $N_{r,k}$. Every coordinate of a point of $\mathcal{W}_n$ is an integer multiple of $1/n$. Therefore distinct points of $\mathcal{W}_n$ have ℓ^∞ distance at least $1/n$. Shrinking $\delta^0_{r,k}$ below $1/(2N_{r,k})$ makes

$$\omega_{n,r}(2\delta^0_{r,k}) = 0, \qquad n < N_{r,k}.$$

After a further deterministic shrinking, we may arrange that

$$\delta_{r,k} \downarrow 0 \quad \text{as } k \to \infty$$

and

$$\sup_n \mathbb{P}[\omega_{n,r}(2\delta_{r,k}) > k^{-1}] \le \eta 2^{-r-k-3}. \tag{15}$$

Let E_n be the event that simultaneously, for every $r, k \in \mathbb{N}$,

$$H_{n,r} \le A_r \quad \text{and} \quad \omega_{n,r}(2\delta_{r,k}) \le k^{-1}.$$

The union bound, (14), and (15) give

$$\inf_n \mathbb{P}(E_n) \ge 1 - \eta.$$

Let $\mathscr{K}_\eta$ be the closure of all realizations of $(\mathcal{W}_n, \Gamma_n)$ which satisfy E_n. By Theorem 7.1, $\mathscr{K}_\eta$ is compact and every element of $\mathscr{K}_\eta$ is a continuous graph over its first component. Moreover,

$$\inf_n \mathbb{P}[(\mathcal{W}_n, \Gamma_n) \in \mathscr{K}_\eta] \ge 1 - \eta.$$

This proves tightness.

Let μ be any subsequential weak limit. Since $\mathscr{K}_\eta$ is closed, the closed-set half of the Portmanteau theorem gives

$$\mu(\mathscr{K}_\eta) \ge \limsup_n \mathbb{P}[(\mathcal{W}_n, \Gamma_n) \in \mathscr{K}_\eta] \ge 1 - \eta.$$

Every element of $\mathscr{K}_\eta$ is a locally bounded continuous graph. Since $\eta > 0$ is arbitrary, μ is supported on such graphs. This proves the theorem. □

Proof of Corollary 1.2. The first coordinate of any joint subsequential limit has the prescribed law of $\mathcal{W}_\infty$. Theorem 1.1 says that its second coordinate is almost surely a continuous graph over that first coordinate. □

9 Adversarial verification report

The following table records the defects found in the earlier response and the corresponding repairs. "Fatal" means that the earlier text did not justify the stated conclusion; it does not mean that the theorem itself is false.

Severity	Earlier defect	Repair in this note
Fatal	The second Baxter coordinate was treated as another i.i.d. KMSW order without proof.	Section 1.1 cites the precise primal-west/dual-west identification and the equal-law KMSW encoding from Gwynne–Holden–Sun, Remark 1.8 and Lemma 2.5.
Fatal	The proof switched from the blackboard's edge-time domain to white vertex corners and claimed the difference was harmless without defining it.	The entire theorem is now stated directly on edge points $p(e) = (i(e), j(e))$ and uses closed Baxter rectangles. Lemma 2.1 proves the exact edge-path formula.
Fatal	It asserted convergence of the discrete equality relation to the continuum equality set merely from peanosphere convergence.	That claim is removed. Remark 1.3 separates distance-field tightness from exact meeting-relation convergence; Corollary 1.2 gives the correct implication once a domain theorem is supplied.
Major	Tightness of $H_{n,R}$ was incorrectly converted into almost-sure boundedness on a Skorokhod representation.	Sections 7–8 replace this by deterministic compact containment sets and a variable-domain Arzelà–Ascoli theorem.
Major	The singleton-fiber argument invoked an unproved openness assertion for oscillation events.	Theorem 7.1 proves projection equality, singleton fibers, and continuity directly for every deterministic closed-set limit.
Major	The local Hausdorff topology was not specified, and ordinary intersections with truncating boxes can behave badly at box boundaries.	Section 1.2 uses the Fell topology, equivalently Hausdorff convergence in the one-point compactification.
Major	The block variable was not distinguished carefully from possible completion edges.	Lemma 2.3 works with the labelled active-edge incidence graph and cites the exact KMSW restriction theorem. Only edges in the packet are used.
Minor	Several endpoint conventions and m versus $m+1$ counts were suppressed.	The active edges are explicitly $\lambda_0, \dots, \lambda_m$; all estimates use $(m+1)^{3/4}$ and strip cardinalities include harmless $+3$ terms.

9.1 Dependency audit

The proof uses the following external inputs and no others.

1. **BBMF Hasse representation.** Black points represent edges; white points represent vertices; every black point has one lower and one upper white neighbor; suppressing black points yields the bipolar orientation [1].
2. **KMSW interval consistency.** Restricting a walk to a finite consecutive interval gives the

corresponding marked submap with inherited incidences [2, Proposition 2.4].

3. **Second Peano order.** The primal west-going encoding is an i.i.d. three-step walk with the same law as the usual encoding, and its edge order agrees with the dual west-going order [4, Remark 1.8 and Lemma 2.5].
4. **One upper quantile at exponent** $3/4$**.** Borga–Gwynne Theorem 1.12 supplies a uniform constant-order upper bound for the maximal LDP in an m-step KMSW cell [3].
5. **Reimer's inequality.** Arbitrary events in a finite product space satisfy the disjoint-occurrence inequality [6].

Everything after these inputs—the closed-rectangle formula in the endpoint form used here, the zero-buffer recursion, the exponential tail, the four-strip inequality, and the variable-domain compactness theorem—is proved in this note.

10 Exhaustive finite-state audit

A separate Python program reconstructs the directed active-edge incidence graph from the exact KMSW endpoint identities. It performs the following checks.

1. For every three-step word through length eight, head-to-tail incidences respect strict increase of exploration labels.
2. For every such word and every consecutive label interval, the complete signed-endpoint equality matrix agrees with the equality matrix of the freshly run subword. This checks the interval isomorphism without using the proof text.
3. For each word length through eight, it computes the exact distribution of Δ_m over all 3^m words and verifies, for every admissible r, k, the integer form of
$$\mathbb{P}[\Delta_m \geq kr] \leq \mathbb{P}[\Delta_m \geq r]^k.$$

The output was:

```
PASS: endpoint monotonicity and interval isomorphism
  words checked:  9841
  word-interval pairs checked:  401041
PASS: exact zero-buffer BKR tail inequalities
  all thresholds r,k checked for m = 0,1,...,8.
```

The exact diameter histograms were

m	#{words with each diameter $d = 1, 2, \ldots$}
0	(1)
1	$(2, 1)$
2	$(4, 4, 1)$
3	$(8, 12, 6, 1)$
4	$(16, 33, 23, 8, 1)$
5	$(32, 88, 74, 38, 10, 1)$
6	$(64, 231, 221, 143, 57, 12, 1)$
7	$(128, 599, 638, 479, 248, 80, 14, 1)$
8	$(256, 1537, 1811, 1506, 930, 397, 107, 16, 1)$.

These checks are evidence against implementation-level mistakes in the deterministic KMSW portion. They are not a substitute for the proofs above and are not described as formal verification.

11 Conclusion

The corrected argument establishes the actual graph-tightness statement. The essential estimate is

$$\mathbb{P}[\Delta_m > A(m+1)^{3/4}] \le Ce^{-cA},$$

which follows from one known $m^{3/4}$ upper quantile and the zero-buffer BKR recursion. This exponential tail survives the union bound over all $O(\delta^{-1})$ narrow intervals in a macroscopic window. The two Peano orders then control the four sides of every Baxter rectangle, giving the endpoint modulus

$$\lim_{\delta \downarrow 0} \limsup_{n} \mathbb{P}[\omega_{n,R}(\delta) > \varepsilon] = 0.$$

The variable-domain compactness theorem converts this modulus and the local height bound into the claimed result: every subsequential limit is a graph, every vertical fiber is a singleton, and the limiting function is continuous.

References

[1] N. Bonichon, M. Bousquet-Mélou, and É. Fusy, *Baxter permutations and plane bipolar orientations*, Séminaire Lotharingien de Combinatoire 61A (2010), Art. B61Ah. arXiv:0805.4180.

[2] J. Borga and M. Maazoun, *Scaling and local limits of Baxter permutations and bipolar orientations through coalescent-walk processes*, Annals of Probability 50 (2022), no. 4, 1359–1417. arXiv:2008.09086.

[3] J. Borga and E. Gwynne, *Directed distances in bipolar-oriented triangulations: exact exponents and scaling limits*, preprint (2025). arXiv:2510.26123.

[4] E. Gwynne, N. Holden, and X. Sun, *Joint scaling limit of a bipolar-oriented triangulation and its dual in the peanosphere sense*, Acta Mathematica Sinica, English Series (2026). arXiv:1603.01194.

[5] R. Kenyon, J. Miller, S. Sheffield, and D. B. Wilson, *Bipolar orientations on planar maps and* SLE_{12}, Annals of Probability 47 (2019), no. 3, 1240–1269. arXiv:1511.04068.

[6] D. Reimer, *Proof of the van den Berg–Kesten conjecture*, Combinatorics, Probability and Computing 9 (2000), no. 1, 27–32.